SDC 4: Characteristics of Included AI Models on Artificial Intelligence in Plastic Surgery

Aesthetic and Breast Surgery

Author, Year	Study Type	Study Phase	AI Subfield	Input Data (D)	Input Data	Data Source	nTrain	nTest	nValidate	Training Ratio	Output Data (D)	Output Data	Comparison	Ground Truth	Accuracy
Xie, 2007[27]	Algorithm development and performance evaluation	0	USML	4	3D skull: 3 points to skull surface and 3 points for implant surface over time	Reconstructed 3D implant model skulls	NR	NR	NR	NR	3	Simulation of collision detection for nose augmentation surgery	O9n^2	NR	NR
Talbar, 2019[35]	Algorithm development and performance evaluation	0	ANN	2	Facial 2D images preop and postop with ALGOH feature extraction (3 layers of granularity)	Patient images from 68 blepharoplasty, 51 brow lift, 51 liposhaving, 15 malar augmentation, 18 mentoplasty, 52 ostoplasty, 75 rhinoplasty, 72 rhytidectomy, 50 skin peeling	452	NR	NR	NR	1	Recognition rate of facial features to detect identity	GLOH algorithm, SIFT and combinations of CSLBP, SIFT, VSIFT, LDA, PCA, CLBP, NN-LM and EUCLBP, and PCA.	True Identity	99.00
Gunes, 2006[70]	In silico algorithm performance optimization	1	SML	2	Facial 2D images, females	Female face images of varying ethnicity and age from public databases: SCUT-FBP, M2B, and SCUT-FBP5500	165	50	NR	77:23	1	Quantitative measurements (attributes) of beauty and classification of facial images.	Cross validation	48 Human referees	92.00
Silva, 2019[30]	In silico algorithm performance optimization	1	ANN	2	Breast reconstruction 2D images (pixels)	ImageNet. VIENNA (non-standardized) and PORTO datasets (standardized)	120	221	NR	65:35	2	Key points of breast photographs after breast surgery	Cross validation	Heatmap and measured 74 key point coordinates of breast contour	99.98
Borsting, 2020[31]	In silico algorithm performance optimization	1	CNN	2	Facial 2D images preop and postop rhinoplasty (pixels)	Rhinonet database	18149	2268	2268	80:10:10	1	Predicted rhinoplasty status	Plastic surgery attendings and residents	Plastic surgery attendings and residents	85.00
Geras, 2017[28]	Algorithm performance in a medical setting	2	CNN	2	2D Mammographic images (2600x2000 pixels)	Large-scale data set of 201,698 screening mammographic exams (886,437 images) collected at multiple sites of authors institution, this was done to use high-resolution images without the need for downscaling. Cropped all pictures to unify the sizes and scale.	709150	88643 .7	88644	80:10	1	BIRADS score	Different training sizes, ensemble	4 Human expert radiologists	78.70
Guan, 2019[29]	Quality Improvement	2	GAN	2	2D Mammographic images with labeled ROIs	DDSM website: PNG pictures of 2620 mammographic images. 695 normal mammograms, 1925 abnormal mammograms (914 malignant/cancers, 870 benign, and 141 benign without callback).	420	336	84	55:45	2	Synthetic mammographic images	Minimum rectangle-shape areas surrounding the whole given ground-truth boundaries.	Minimum rectangle-shape areas surrounding the whole given ground-truth boundaries.	85.12
Patcas, 2019[32]	Algorithm performance in a medical setting	2	CNN	2	Facial 2D images postop aesthetic jaw surgery, annotated with patient data (modified to 256 x 256 pixels)	Internet Movie Database and Wikipedia (IMDb-Wiki, age range: 0–100 years); labeled with apparent age years (age range: 0–95 years); Chicago Face Dataset (CFD)	500000	NR	2164	99.57:.4	1	Apparent facial age and attractiveness (0: extremely unattractive; 100: extremely attractive)	Preoperative age assessment	Real age	NR
O'Neill, 2020[25]	Algorithm performance in a medical setting	2	USML	1	Characteristics of patients (age, body mass index, comorbidities, smoking) procedures (timing and laterality of reconstruction, history of radiation)	1012 patients undergoing DIEP flaps: 299 nonevents (no flap failure) and 269 events of flap failure	568	568	NR	50:50	1	Total flap failure	Standard resampling: random under sampling without replacement, random oversampling with replacement, synthetic minority oversampling technique; random oversampling examples; ensemble machine learning technique	Ultimate flap failure	95.00
Dorfman, 2020[33]	Algorithm performance in a medical setting	2	CNN	2	Facial 2D images after landmarking 68 points (pixels)	Microsoft Azure Face API (Redmond, WA)	890	2667	100	25:75	1	Predicted age	None: would need crowd sourcing	None: would need crowd sourcing	NR

Chen, 2020[34]	Algorithm performance in a medical setting	2	CNN	2	Facial 2D images preop and postop facial feminization.	Dataset of 20 consecutive male-to-female patients from 2013-2018 after all stages of facial feminization surgery performed by two surgeons at a single institution. Included both hard- and soft-tissue facial feminization surgery procedures.	120	NR	NR	NR	1	Identification of gender vs. misidentification (improvement in gender identification represents confidence of femininity.)	(1) Microsoft, (2) IBM, (3) Amazon, and (4) Face++ (5) ground truth	Automated classifier using the average human grades as the ground truth and used it to classify an independent test set of facial images	98.00
Mavioso, 2020[26]	Clinical Trial	3	CV	2	2D preoperative CTs Angiographic images (0.8 mm by 0.4 mm) of patients to undergo DIEP flaps.	Angio-CTs of 40 breast reconstruction patients with 180 perforators.	21	NR	NR	NR	2	Identification of perforators (mapping)	Preoperative manual identification of perforators by imaging team	Calibers of 234 vessels confirmed by the surgical team intra-operatively	75.00
Craniofacial Surgery														
Xi, 2010[80]	Algorithm development and performance evaluation	0	ANN	2	8-10 factors as input data: features defined by degree of deformity. normal/affected nostril: area/center point/angle normal nostril with nasal septum as symmetrical axis (area/center/angle); overlapping area of normal nostril with nasal septum as symmetrical axis and area of affected nostril; distance between center points: normal and affected nostril, affected nostril & desired symmetrical location, normal nostril, and desired symmetrical location of affected one	NR	85	45	40	83:2.3	2	Evaluation score of cleft-lip	Regression Analysis	5 craniofacial and professional cleft surgeons and 5 ordinary persons	92.15
Mendoza, 2014[40]	Algorithm development and performance evaluation	0	SML	3	3D volumetric representations of craniums (voxels) represented as surface models. Anatomically landmarked.	CT images from 123 cases of non-syndromic craniosynostosis	123	NR	141	47:53	1	Classification of craniosynostosis: normal, sagittal, metopic, coronal, true class, assigned class	LDA: linear discriminant analysis, RF: random forests, SVM: support vector machines. MS: mean shape approach; CP: constrained projection approach; linear discriminant analysis; closest normal; operator manually labeled cranial bones; reported accuracy of trained radiologists	Reported accuracy of trained radiologists	95.70
Wu, 2016[36]	Algorithm development and performance evaluation	0	SML	3	3D stereophotogrammetry images	3D images of subjects: 35 with unilateral CL, 10 with bilateral CL, five controls	50	NR	NR	NR	2	Approximation of best midfacial reference place for measuring facial symmetry	Several mid-facial plane applications: 3 computer-based (mirror, deformation, learning), human based (direct placement, manual landmark)	Three cleft surgeons, two craniofacial pediatricians, and one craniofacial researcher.	98.32
Ferry, 2019[39]	Prototype design	0	USML	2	Facial 2D images	Database of facial images, publicly or scientifically published images (100-283 images/ syndrome) converted to JPEG and submitted to the Facial Detection module.	2754	NR	NR	NR	1	List of differential diagnosis for the most likely disorders	NR	Manual annotation of 36 feature points	60.00
Nishimoto , 2019[42]	Algorithm development and performance evaluation	0	CNN	3	3D cephalograms with manually plotted cephalometric landmarks' coordinate values (resized to 600x600 pixels)	219 images of lateral cephalograms from the internet. Training images were expanded 51 fold.	7803	66	NR	99.2:.8	2	Prediction errors for automated landmark predicting system for evaluating craniomaxillofacial skeletal profile	Manually plotted cephalometric landmarks	NR	99.92

Bhalodia, 2020[38]	Algorithm development and performance evaluation	0	SML	2	2D CT images (64 slices, 0.25mm cut)	17 metopic skull CT images and 65 nonaffected skull CT's: 3D skull input into ShapeWorks software, mapped with thousands of different points, and statistically analyzed in several different planes to generate a shape descriptor (xi) which identifies the geometric variations in each skull. Each scan was represented using 8 scalar values to form the proposed shape descriptor.	82	NR	NR	NR	1	Predict severity of metopic craniosynostosis	Traditional IFA model	Craniofacial surgeons (10 craniofacial plastic surgeons and 8 pediatric) neurosurgeons	95.00
Ma, 2020[43]	Algorithm development and performance evaluation	0	CNN	3	3D volumetric CT images (330 slices, 0.351 mm, width, and height of 512 × 512 pixel; voxels)	OMS patient database during 2008–2017 at The University of Tokyo Hospital (66 scans)	58	NR	8	NR	3	Automatic landmarking of CT images.	MIMICS: manual landmarking skull (1. Nasion, 2. Zygomatic point left, 3. Zygomatic point right), maxilla (4. Anterior nasal spine, 5. Incisors intersecting point), mandible (6. Pogonion, 7. Menton), and ramus (8. Gonion left, 9. Gonion right, 10. Ramus notch left, 11. Ramus notch right, 12. Coronoid process left, 13. Coronoid process right)	NR	100.00
Maier, 2008[78]	In silico algorithm performance optimization	1	SML	1	Speech data manually transcribed (7647 phones, 1916 words)	NR	7647	NR	NR	NR	1	NR	Leave-one-speaker-out evaluation	NR	75.80
Sari, 2008[37]	Usability test	1	NLP	1	Phonemes (sustained, nonsense consonant-vowel-consonant-vowel (CVCV) combinations, object names, and short sentences)	Multiple-speaker speech recognition database collected at Sabancı University (Erdogan et, 2005) and using the Hidden Markov Toolkit (Woodland et, 1994). The recordings were force-aligned to the corresponding text transcriptions	NR	NR	NR	NR	1	Recession in vowels, recession for phonemes	NR	NR	NR
Dusseldorp, 2019[41]	Algorithm performance in a medical setting	2	CV	2	Facial 2D images of preop and postop facial reanimation	Affdex: Affectiva, Boston, Mass. 10 normal subjects, 30 facial palsy patients, 134 smile reanimation patients, 61 preoperative patients	NR	NR	225	NR	1	Emotionality quotient (joy, contempt, disgust)	Oral commissure excursion	593 layperson assessments of disfigurement	NR
Non-breast Microsurgery															
Fichera, 2013[48]	Prototype design	0	Combo	2	Laser interactions (change in temperature due to heat transfer & transport - modeled f(laser exposure time))	Monte Carlo-based simulation of laser propagation inside the tissue plus a finite difference method of heat transportation (200 laser interactions, 627120 samples)	627120	NR	176880	78:22	2	Tissue carbonization and incision quality (targets to be automatically supervised)	Training data model, validation model. learned model: continuous laser mode, and pulsed laser mode	Learning data set, actual temperature for volume	96.30
Tatinati, 2015[47]	Prototype design	0	USML	3	Tremor signal (Hz): accelerometer input for sensing tip acceleration in x-, y-, and z-axes separately.	NR	500	NR	NR	NR	2	Prediction of tremor and Hz required for compensation	Single-step prediction model for tremor prediction and multi-step (MS-AR-KF)	NR	71.00
Power, 2018[49]	Prototype design	0	ANN	3	Spectra, corresponding force readings, at a given point in time	NR	400	NR	NR	NR	2	Estimated real-time measurements of force	NR	NR	97.30

Nakazawa, 2020[45]	Prototype design	0	CNN	2	Intraoperative video frames (Recorded in 30 Hz)	Microscopic images from video frames, and pre-trained ImageNet	27	NR	NR	NR	3	Features and regional proposals	NR	Bounding box annotated by an engineering student for the left images with (460 × 540 pixels)	89.20
Kiranantawa, 2014[46]	In silico algorithm performance optimization	1	SML	2	2D quality-controlled images of free flaps	Galaxy S2 smartphone images (20 subjects, 60 images, minimum 0.5 million pixels). Quality of images controlled for predetermined light intensity, camera-to-object distance, and white balance.	60	NR	126	32:68	1	(1) normal, (2) partial venous occlusion area, (3) complete venous occlusion area, (4) partial arterial occlusion area, or (5) complete arterial occlusion area. diagnostic postoperative microsurgery monitoring	NR	NR	95.00
Kuo, 2018[44]	Algorithm performance in a medical setting	2	ANN	1	Patient characteristics (50 inputs including demographics, comorbidities, laboratory values, treatment received)	Patient characteristics collected from preoperative EHR data of 1,854 free-flap reconstructions and 438 postoperative SSIs.	1298	538	NR	7:3	1	prediction of free flap-post-op SSI	Logistic Regression	Actual flap failure	88.37
Burn Surgery															
Estahbanati, 2002[50]	Algorithm development and performance evaluation	0	ANN	1	Patient characteristics (age, sex, TBSA, data of admission, lapse time, refereed or non-refereed status, inhalation injury, hematology, biochemistry lab values, medical outcome, number, type of surgical episodes (debridement or skin graft)	Input data points were obtained during a 1-year period from medical files from Motahary Hospital (a major referral burn center in Tehran)	811	271	NR	75:25	1	Survival	NR	Expert physicians	90.00
Yeong, 2005[52]	Algorithm development and performance evaluation	0	ANN	1	Burn Spectra per wound (nm)	47 wounds of 39 patients with less than 20% TBSA burns-reflected burn spectra	41	NR	NR	NR	1	Deep burns (healing >14 days) vs. superficial burns (healing <14 days)	Historically reported accuracy of direct visualization (50-75%); laser doppler flowmetry (90%)	True survival	96.00
Heredia-Juesas, 2016[54]	Algorithm development and performance evaluation	0	SML	2	Feature input from burn cases. Features grouped by measurements: photoplethysmography (PPG) (blood flow in skin's microcirculation) (14 features), real image (RI) features (black and white photo; 11 features), multi-spectral imaging (MSI) tissue reflectance spectrum at key wavelengths (11 features)	Cases of 60 burns (6 lesions, 5 stages of injury, 3 features)	60	NR	NR	NR	1	Classification of burn: healthy skin, deep burn, shallow burn	Cross-validation	NR	77.00
Yamamura, 2007[51]	In silico algorithm performance optimization	1	ANN	1	Patient characteristics (burn area after skin graft operation (%), blood sugar (mg/dL), duration of dose (day), serum creatinine (mg/dL), Cmax of ABK)	EHR collected wound assessment data of 25 burn patients	NR	25	NR	NR	1	ABK response ‘‘excellent’’= 3, ‘‘very good’’= 2, or ‘‘good’’ =1	Leave-one-out cross-validation.	NR	100.00

Martínez-Jiménez, 2018[53]	In silico algorithm performance optimization	1	SML	1	Infrared thermography (difference between temperature of the burn and the surrounding 25 cm2; ΔT)	Patients with partial or full thickness burns who were admitted to the burn care unit within 24 hours from injury.	34	100	22	25:75	1	Prediction of treatment modality of either healing by re-epithelization, requiring skin grafts, or requiring amputations	Inter-rater agreement weighted kappa coefficient	Treatment modality	85.35
General Plastic and Reconstructive Surgery															
Sarker, 2015[57]	In silico algorithm performance optimization	1	NLP	1	1132 evidence-based answers from 2713 medical documents. Preprocessed.	In-house developed corpus for specialized EBM question answering using 'Clinical Inquiries' from the Journal of Family Practice.	1132	NR	NR	NR	1	NR	Human comparison	Human comparison	63.84
Baur, 2018[23]	In silico algorithm performance optimization	1	GAN	2	2D Dermatoscopic images	ISIC2018 dataset: 10,000 labeled training samples	10000	NR	NR	NR	1	NR	Visual Turing Test (VTT) involving 3 expert dermatologists (ED) and 5 deep-learning experts (DLE)	Visual Turing Test (VTT) involving 3 expert dermatologists (ED) and 5 deep-learning experts (DLE)	50.00
Levites, 2019[55]	Usability test	1	NLP	1	Short-form textual data	Twitter Mentions, Cognovi Labs AI	NR	NR	249687	NR	1	Measurements of awareness, engagement, motivation	NR	NR	NR
Boczar, 2020[56]	Usability test	1	NLP	1	10 frequently asked topics in plastic surgery, 30 participants.	IBM Watson Assistant online platform for training AIVA chatbots	10	300	300	NR	1	Responses to participants	Correct answer for participant questions	Correct answer for participant questions	92.3
Oral & Maxillofacial Surgery															
Knoops, 2019[60]	Algorithm development and performance evaluation	0	Combo	3	3D facial scans	Large Scale Facial Model database (LSFM): 9663 scans: 4,261 faces (healthy volunteers and orthognathic surgery patients)	3408.80	852.20	NR	80:20	1	Automated diagnosis (classification) of faces with orthognathic shape features as an indication for orthognathic surgery; simulate (mean surgical correction measurements) patient faces for automated patient-specific surgical planning	Compared compactness, generalization, and specificity of three models: global model, bespoke preoperative model to LSFM	Real faces; monte-Carlo cross-validation (1000x)	95.40
Zhang, 2018[58]	Algorithm performance in a medical setting	2	ANN	1	Patient characteristics (gender, age, physique, blood pressure, oral hygiene, relation of wisdom teeth, molars, mandible, dental arch, incision type, quality of bone removal, root fracture condition, lingual bone fracture, surgical time).	300 patients used as training sample	300	NR	100	3:1	1	Prediction of swelling after impacted third mandibular molar extraction	5-fold cross validation	Patient outcomes	98.00
Stehrer, 2019[59]	Algorithm performance in a medical setting	2	SML	1	Patient characteristics	Patients undergoing orthognathic surgery	760	190.00	760	80:20	1	Perioperative blood loss	Linear regression	Actual values for perioperative blood	98.24
Wound Healing															
Robnik-Sikonja, 2003[79]	Algorithm performance in a medical setting	2	SML	1	Wound Measurements (mm/day)	214 patients, 300 wounds	300	NR	NR	NR	1	Classification: non-healing wounds, slow healing wounds, medium healing wounds, fast healing wounds	NR	NR	80.00

Jung, 2016[61]	Algorithm performance in a medical setting	2	SML	1	Patient characteristics (865 predictors): demographic information, ICD9 codes, wound measurements and categorical characteristics.	Healogics wound care centers 68 in 26 states from 2009 to 2013. 59,953 patients, 150277 wounds.	120473	29804	NR	80:20	1	Probabilities of delayed wound healing	Standard of care	Patient outcomes	84.20
Hand Surgery															
Luján, 2004[64]	Prototype design	0	ANN	2	Preprogrammed External Stimulations (pulse widths)	Stimulation for Muscles: adductor pollicis, brachioradialis, extensor carpi ulnaxis extensor carpi radialis and brevis, extensor pollicis, flexor digitorium superficialis, flexor pollicis longus	50	NR	NR	NR	2	Outputs of the musculoskeletal system: actual forces and angles. Stimulus generated by the computer system used to provide independent control of hand grasp and wrist posture.	NR	NR	NR
Hincapie, 2009[68]	Algorithm development and performance evaluation	0	ANN	2	Voluntary Activated Movement Recordings (kinematics)	Movements (3 subjects, 418 movements) from input muscles: trapezius (clavicular and scapular), deltoid (clavicular and scapular), rhomboids, biceps, subscapularis, supinator	250.8	83.6	83.6	6:4	2	Prediction of functional electrical stimulation	Predict necessary functional electrical stimulation for muscles typically paralyzed in C5 SCI (infraspinatus, serratus anterior, triceps)	NR	96.4

Table 4: Characteristics of Artificial Intelligence Models of Included Articles. Articles within this table are divided by plastic surgery subspecialty and study phase, sorted chronologically by year. The following qualitative and quantitative characteristics of the included articles are described as follows. The study types as described by Park et al. and modified for this paper include algorithm development and performance evaluation and prototype design (phase 0), prototype design (phase 0), in silico algorithm performance optimization (phase 1), usability test (phase 1), interface design (phase 2), quality improvement (phase 2), clinical trial (phase 3), adverse events identification (phase 3), post deployment surveillance (phase 4). AI subfields include Supervised Machine Learning (SML), Computer Vision (CV), Natural Language Processing (NLP), Unsupervised Machine Learning (USML), Artificial Neural Network (ANN), Convolutional Neural Network (CNN). Input data dimension: one-dimensional (1D), two-dimensional (2D), three-dimensional (3D), four-dimensional (4D). Output data dimension: one-dimensional (1D), two-dimensional (2D), three-dimensional (3D), four-dimensional (4D). Number of data used for training (nTrain), testing (nTest), validating (nValidate). Training ratio: the ratio of training to testing data. The comparison was considered "equivalent" if it was like the study group as measured by criteria other than the study endpoints and if there were no clear confounding variables that could influence the comparison. The study was considered to have an "adequate control group" if it was compared to a gold standard diagnostic test or therapeutic intervention recognized as the optimal intervention according to the available published data. The ground truth was considered contemporary if the control and study groups were managed during the same time and there was no historical comparison. Performance accuracy: the highest reported probability as reported in the body of the included manuscript.